DOES AIFS NEED POST-PROCESSING? STATION FORECASTS FOR TUNISIA COMPARED WITH IFS AND GFS

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ABSTRACT

We correct daily maximum and minimum temperature and rain forecasts from ECMWF IFS, ECMWF AIFS and NCEP GFS at 12 Tunisian SYNOP stations, 1 to 7 days ahead. Temperature is corrected with quantile LightGBM plus conformal intervals, rain with a calibrated two-stage model. Models are retrained every month and tested on September 2025 to October 2026. At day 1 the $T_{\max}$ mean absolute error (MAE) drops by 49% for IFS, 50% for AIFS and 39% for GFS (63% at Sfax), and the gain is significant at every lead. After correction AIFS has the lowest error (1.21 °C averaged over leads). Most of the gain, though, is bias removal: LightGBM is 2–4% better than linear MOS for IFS and GFS and not significantly better for AIFS. Conformal recalibration raises the coverage of the 80% intervals from 69–78% to 80–84%. Rain probabilities reach a Brier skill score of about 0.2 against climatology, while the raw forecasts are below climatology. A GRU did not improve on LightGBM.

Keywords post-processing · model output statistics · AIFS · LightGBM · conformal prediction · Tunisia

1 Introduction

The ECMWF IFS and NCEP GFS fields used in this study are on a 0.25° grid (about 25 km). At that resolution the Tunisian coastline and the dry interior are only roughly represented, and the error at a given station is partly systematic. Model Output Statistics (MOS) removes this kind of error with regressions on past forecast–observation pairs (Glahn and Lowry, 1972). Since then many statistical and machine-learning methods have been used for the same purpose (Vannitsem et al., 2021), including neural networks (Rasp and Lerch, 2018). ECMWF now also runs a machine-learned global model, AIFS (Lang et al., 2024), and it is not obvious whether its station errors behave like those of the physical models.

We look at three questions:

1. How much does a gradient-boosted MOS reduce the temperature errors of IFS, AIFS and GFS at leads 1–7?
2. Are the resulting 80% intervals and rain probabilities calibrated?
3. Which inputs do the corrections depend on?

All inputs are freely available and the code is public.

2 Data

2.1 Observations

We use SYNOP reports from the 12 stations in Table 1. Reports up to 2025-08-24 come from NOAA Global-Hourly (Smith et al., 2011). NOAA has no Tunisian reports after that date, so later reports are taken from Ogimet and decoded

Table 1: Stations, ordered from north to south, and number of valid observations between 2025-09-01 and 2026-10-25.

					Days in test period		
Station	WMO	Lat.	Lon.	Elev. (m)	$T_{\max}$	$T_{\min}$	Rain
Bizerte	60714	37.24	9.79	6	400	396	409
Tabarka	60710	36.98	8.88	70	413	420	417
Tunis	60715	36.85	10.23	7	405	404	404
Kelibia	60720	36.85	11.08	30	392	394	393
Jendouba	60725	36.48	8.80	144	410	398	348
Enfidha	60731	36.08	10.43	6	400	395	392
Kairouan	60735	35.67	10.10	68	409	409	415
Sfax	60750	34.72	10.69	26	405	385	387
Gafsa	60745	34.42	8.82	323	405	277	287
Tozeur	60760	33.94	8.11	87	407	381	376
Gabès	60765	33.88	10.10	8	411	419	419
Djerba	60769	33.88	10.78	6	403	378	373

with our own parser. The two sources give identical values on the 66–84 days per station where they overlap. Daily values follow the SYNOP conventions (all times UTC): $T_{\max}(D)$ is the maximum over 06–18 on day D, $T_{\min}(D)$ the minimum over the 24 h ending at 06 on D, and $P(D)$ the rain from 06 on D to 06 on $D{+}1$. A day with $P \geq 1$ mm is a wet day. We did not use Meteostat, because its rain values for these stations come from the DWD MOSMIX forecast and not from measurements.

Quality control sets a value to missing if it is out of range, if $T_{\min} > T_{\max}$, if it jumps by more than 12 °C and back, if it repeats for five days, or if it is more than six robust standard deviations from the station's monthly median. This removed 48 of about 160 000 values. Gabès has SYNOP reports only from 2019, and Gafsa has many missing night reports (Table 1).

2.2 Forecasts

Archived hourly forecasts come from the Open-Meteo Previous-Runs API (Zippenfenig, 2023) for IFS and GFS (from February 2024) and AIFS (from February 2025). We use 2 m temperature, rain, 2 m dew point, total cloud cover and 10 m wind. For a forecast at lead L valid on day D, every hour is taken from the run that was available at the end of $I = D - L$, and the daily windows are the same as for the observations. The archive goes back 7 days, and rain over 06–06 needs one extra day, so rain is only available up to lead 6.

3 Methods

3.1 Predictors

A forecast for day D at lead L is treated as issued at the end of day $I = D - L$. It may use $T_{\max}(I)$, $T_{\min}(I)$ and $P(I-1)$ (since $P(I)$ is only complete at 06 UTC on $I{+}1$), and forecast errors up to I. The 17 temperature predictors are the five forecast fields, lead, day of year (sine and cosine), station latitude, longitude and elevation, the last observed $T_{\max}$ and $T_{\min}$, the climatological $T_{\max}$ and $T_{\min}$ (31-day smoothed, 2014–2023), and the mean $T_{\max}$ and $T_{\min}$ forecast errors of the last 30 days. A unit test changes every observation after I and checks that none of the predictors changes.

3.2 Temperature

The baselines are the raw forecast, climatology, persistence of the last observation, the raw forecast minus its 30-day mean error, and linear regression on the same predictors (linear MOS). The main model is LightGBM (Ke et al., 2017) with quantile loss at $\tau = 0.1, 0.5, 0.9$, trained on the error of the raw forecast rather than on the temperature itself, so that the corrected forecast is not limited to values seen in training. One model is fitted per NWP model and target, for all stations and leads together. The 10–90% interval is recalibrated with conformalized quantile regression (Romano et al., 2019), using the out-of-sample scores of the previous 365 days for the same model, target and lead.

Table 2: Temperature MAE (°C) of the raw and LightGBM forecasts, all stations. Gain is the relative MAE reduction with a 95% block-bootstrap interval.

Model	Target	Lead (d)	Days	MAE raw	MAE LightGBM	Gain % [95% CI]
IFS	$T_{\max}$	1	4703	1.74	0.89	48.6 [44.9, 52.3]
IFS	$T_{\max}$	3	4680	1.89	1.05	44.8 [41.0, 48.5]
IFS	$T_{\max}$	5	4680	2.08	1.35	35.3 [31.6, 39.0]
IFS	$T_{\max}$	7	4680	2.61	1.80	31.1 [27.0, 35.2]
IFS	$T_{\min}$	1	4353	1.48	1.02	31.2 [27.8, 34.9]
IFS	$T_{\min}$	3	4355	1.57	1.11	29.3 [26.4, 32.3]
IFS	$T_{\min}$	5	4344	1.71	1.28	25.2 [22.3, 28.1]
IFS	$T_{\min}$	7	4336	1.98	1.54	22.2 [19.3, 25.2]
AIFS	$T_{\max}$	1	4703	1.80	0.90	49.9 [46.8, 53.1]
AIFS	$T_{\max}$	3	4680	1.86	1.02	45.4 [41.7, 49.2]
AIFS	$T_{\max}$	5	4520	2.01	1.24	38.4 [33.9, 42.3]
AIFS	$T_{\max}$	7	4520	2.28	1.61	29.3 [24.4, 34.0]
AIFS	$T_{\min}$	1	4353	1.62	1.08	33.4 [27.9, 38.6]
AIFS	$T_{\min}$	3	4355	1.74	1.16	33.1 [27.5, 38.7]
AIFS	$T_{\min}$	5	4202	1.86	1.32	29.4 [24.2, 35.2]
AIFS	$T_{\min}$	7	4200	1.98	1.47	25.4 [20.6, 30.7]
GFS	$T_{\max}$	1	4703	1.53	0.93	39.4 [35.2, 43.3]
GFS	$T_{\max}$	3	4680	1.64	1.10	33.2 [29.3, 37.1]
GFS	$T_{\max}$	5	4680	1.90	1.43	24.5 [21.2, 28.2]
GFS	$T_{\max}$	7	4680	2.33	1.99	14.7 [10.7, 18.5]
GFS	$T_{\min}$	1	4353	2.03	1.18	41.9 [35.0, 47.9]
GFS	$T_{\min}$	3	4355	2.07	1.28	38.4 [32.2, 44.1]
GFS	$T_{\min}$	5	4344	2.22	1.46	34.4 [28.0, 40.0]
GFS	$T_{\min}$	7	4336	2.32	1.68	27.5 [22.0, 32.7]

3.3 Rain

A LightGBM classifier gives the probability of a wet day and is calibrated with isotonic regression on the last quarter of the training period. A quantile LightGBM trained on wet days only, on $\log(1+P)$, gives the median and 90th percentile of the amount. The baselines are climatology, the raw forecast as a yes/no event ($\geq$1 mm) and logistic regression. For heavy rain ($\geq$10 mm) we issue a warning when the wet-day probability is at least 0.5 and the 90th percentile of the amount is at least 10 mm; the raw forecast warns when it forecasts 10 mm or more.

3.4 Evaluation

For each test month M the models are trained on all rows dated more than 8 days before the start of M, so that no training target is observed after the forecasts in M are issued. The months from March to August 2025 are also predicted, but only to calibrate the conformal intervals. The scored period is 2025-09-01 to 2026-10-25 for every model, with 4 200–4 700 station-days per lead, and all methods are compared on the same rows. LightGBM settings (400 trees, learning rate 0.03, 15 leaves) were fixed beforehand and not tuned. Confidence intervals come from a block bootstrap over calendar weeks with 1000 resamples, which keeps the correlation between nearby days and between stations on the same day. SHAP values are computed with TreeSHAP (Lundberg et al., 2020).

4 Results

4.1 Temperature

The corrected forecasts have a lower MAE than the raw ones for every model, lead and target, and none of the 95% intervals includes zero (Table 2, Figure 1). At lead 1 the $T_{\max}$ MAE goes from 1.74 to 0.89 °C for IFS, from 1.80 to 0.90 for AIFS and from 1.53 to 0.93 for GFS. The reduction gets smaller with lead and is 15–31% at day 7. At Sfax the raw IFS and AIFS $T_{\max}$ errors are larger (2.1–2.2 °C at lead 1), possibly because the sea breeze is not resolved, and the reduction at lead 1 is 63%. The reduction is of similar size in all seasons (30–51% for $T_{\max}$ at leads 1–3).

A 30-day running bias correction already removes most of the error (Table 3). Pooled over leads, LightGBM is better than linear MOS by 2.7–4.0% for IFS (95% CI 1.3–5.9%) and by 2.2–3.3% for GFS. For AIFS the difference is

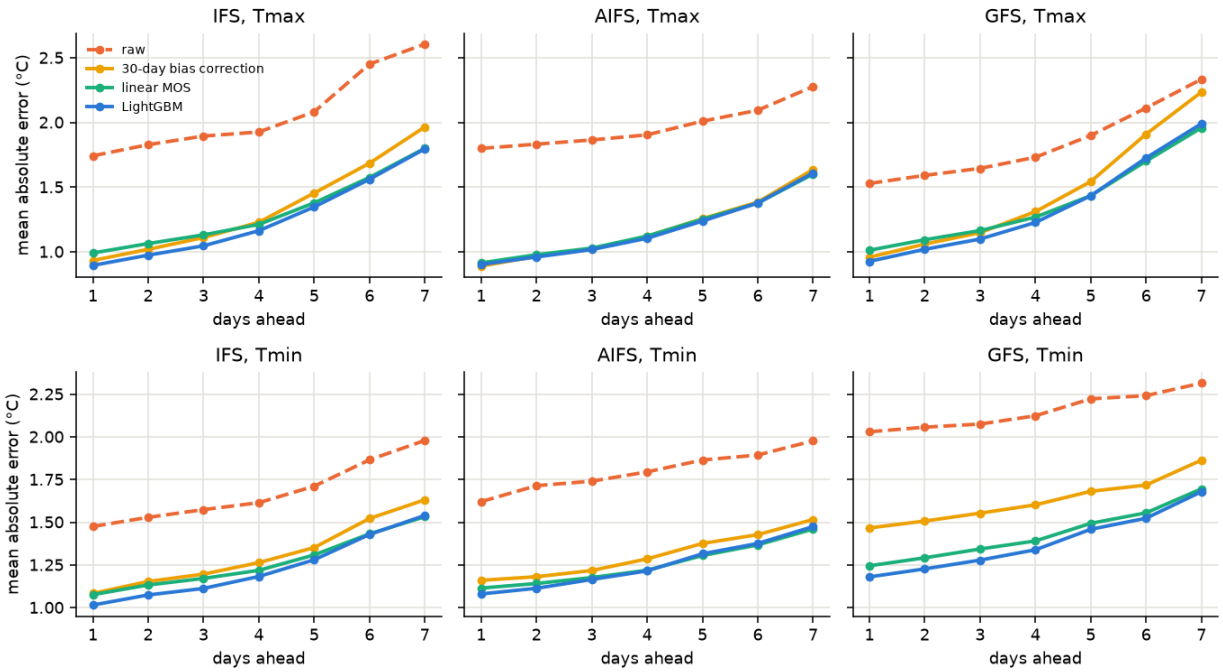


Figure 1: MAE by lead time for the raw forecast (dashed), the 30-day bias correction, linear MOS and LightGBM.

Table 3: MAE (°C) of all temperature methods at leads 1, 4 and 7. The lowest value in each row is in bold.

Model	Target	Lead	Raw	Clim.	Persist.	Bias-corr.	Linear	LightGBM
IFS	$T_{\max}$	1	1.74	2.42	1.97	0.93	0.99	**0.89**
IFS	$T_{\max}$	4	1.93	2.42	3.00	1.23	1.21	**1.16**
IFS	$T_{\max}$	7	2.61	2.42	3.21	1.96	1.80	**1.80**
IFS	$T_{\min}$	1	1.48	2.12	1.82	1.08	1.08	**1.02**
IFS	$T_{\min}$	4	1.61	2.11	2.50	1.26	1.22	**1.18**
IFS	$T_{\min}$	7	1.98	2.12	2.73	1.63	**1.53**	1.54
AIFS	$T_{\max}$	1	1.80	2.42	1.97	**0.89**	0.91	0.90
AIFS	$T_{\max}$	4	1.90	2.42	2.95	1.12	1.12	**1.10**
AIFS	$T_{\max}$	7	2.28	2.42	3.18	1.64	**1.60**	1.61
AIFS	$T_{\min}$	1	1.62	2.12	1.82	1.16	1.11	**1.08**
AIFS	$T_{\min}$	4	1.79	2.13	2.50	1.29	1.22	**1.22**
AIFS	$T_{\min}$	7	1.98	2.13	2.75	1.52	**1.46**	1.47
GFS	$T_{\max}$	1	1.53	2.42	1.97	0.96	1.01	**0.93**
GFS	$T_{\max}$	4	1.73	2.42	3.00	1.31	1.27	**1.23**
GFS	$T_{\max}$	7	2.33	2.42	3.21	2.23	**1.96**	1.99
GFS	$T_{\min}$	1	2.03	2.12	1.82	1.47	1.25	**1.18**
GFS	$T_{\min}$	4	2.12	2.11	2.50	1.60	1.39	**1.34**
GFS	$T_{\min}$	7	2.32	2.12	2.73	1.86	1.70	**1.68**

0.5–0.8% and not significant. Against the 30-day bias correction LightGBM gains 6% for IFS and 7–15% for GFS, but only 0.7% ($T_{\max}$, not significant) and 4.6% ($T_{\min}$) for AIFS, which suggests that the AIFS errors are closer to a constant offset. On the rows where all three models are available the corrected MAE is 1.21 °C for AIFS, 1.24 for IFS and 1.36 for GFS, against 1.89, 1.89 and 1.99 before correction.

4.2 Intervals

The quantile model on its own gives intervals that are too narrow: 69–78% of the observations fall inside the nominal 80% interval. After conformal recalibration the coverage is 80.2–83.5%, with intervals 9–34% wider (Table 4). For

Table 4: Coverage and mean width (°C) of the 80% intervals.

		Quantile LightGBM		+ Conformal	
Model	Target	Cover %	Width	Cover %	Width
IFS	$T_{\max}$	77.5	3.80	81.3	4.13
IFS	$T_{\min}$	76.2	3.66	80.2	4.00
AIFS	$T_{\max}$	73.4	3.25	81.4	3.89
AIFS	$T_{\min}$	69.3	3.25	83.5	4.34
GFS	$T_{\max}$	75.9	3.83	81.2	4.35
GFS	$T_{\min}$	73.4	3.89	80.3	4.49

Table 5: Rain occurrence ($\geq$1 mm), leads 1–6. BSS is the Brier skill score against climatology.

Model	Method	Brier	BSS	ROC AUC	PR AUC
IFS	Climatology	0.103	0.000	0.760	0.349
IFS	Raw	0.136	-0.324	0.772	0.366
IFS	Logistic	0.085	0.177	0.855	0.539
IFS	LightGBM	0.082	0.204	0.867	0.546
AIFS	Climatology	0.103	0.000	0.760	0.350
AIFS	Raw	0.150	-0.453	0.836	0.406
AIFS	Logistic	0.080	0.225	0.871	0.597
AIFS	LightGBM	0.081	0.214	0.884	0.576
GFS	Climatology	0.102	0.000	0.760	0.348
GFS	Raw	0.131	-0.283	0.728	0.332
GFS	Logistic	0.085	0.171	0.854	0.532
GFS	LightGBM	0.083	0.193	0.865	0.535

AIFS $T_{\min}$ the recalibrated interval is somewhat too wide (83.5%), possibly because the AIFS calibration window starts with only a few months of predictions.

4.3 Rain

Used as a yes/no forecast, the raw models have a negative Brier skill score (-0.28 to -0.45), because they forecast rain on too many days. The calibrated LightGBM reaches 0.19–0.21 with a ROC AUC of 0.87–0.88 (Table 5, Figure 2). Logistic regression scores 0.17–0.23, so for rain occurrence most of the benefit comes from calibration and not from the choice of model. Heavy rain is still poorly forecast. Of the 822 station-days with at least 10 mm, the warning rule finds 46–53%, against 26–37% for the raw forecasts, but 73–79% of its warnings are false alarms. On wet days the MAE of the amount is 5.2 and 5.2 mm for IFS and GFS (raw: 6.1 and 6.3 mm), but 5.4 mm for AIFS against 5.3 mm raw.

4.4 What the corrections depend on

For the IFS $T_{\max}$ correction the predictors with the largest mean $|\mathrm{SHAP}|$ are the forecast itself (0.85 °C), the 30-day bias (0.45) and climatology (0.35), followed by wind (0.18) and day of year (Figure 3). High forecasts are corrected downwards and low ones upwards, i.e. towards climatology, and strong forecast wind lowers $T_{\max}$. For $T_{\min}$ the 30-day bias comes first (0.65), then the forecast (0.59), latitude and dew point. For rain the forecast amount (0.91) and cloud cover (0.55) matter most.

The pull towards climatology works against the forecast on extreme days. On 2026-08-23 Sfax reached 45.7 °C. IFS forecast 39.7 °C one day ahead, and the model added only 0.4 °C: compared with the average correction of $+1.25$, the high forecast value contributed -2.0 and the recent bias $+0.6$. The corrected forecast was still 5.6 °C too low. SHAP describes what the model uses, not what causes the errors.

4.5 GRU

A GRU was trained on the same IFS rows, months and target. Its inputs are the observations of the last 14 days plus the 17 LightGBM predictors. Its MAE is within 0.03 °C of LightGBM at all leads ($T_{\max}$ lead 1: 0.887 against 0.898; lead 7: 1.769 against 1.794). The longer observation history therefore adds little to the 30-day bias predictor.

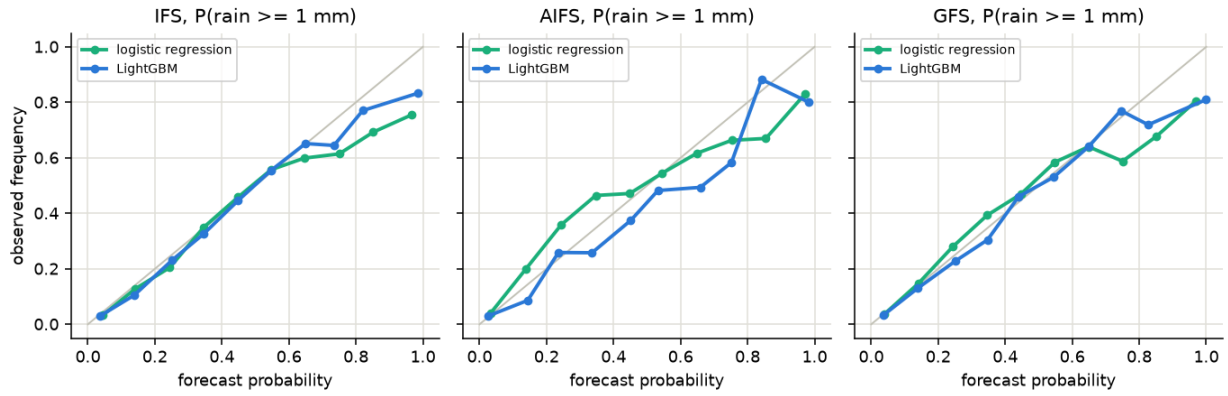


Figure 2: Reliability of the wet-day probabilities. Bins with fewer than 10 cases are not shown.

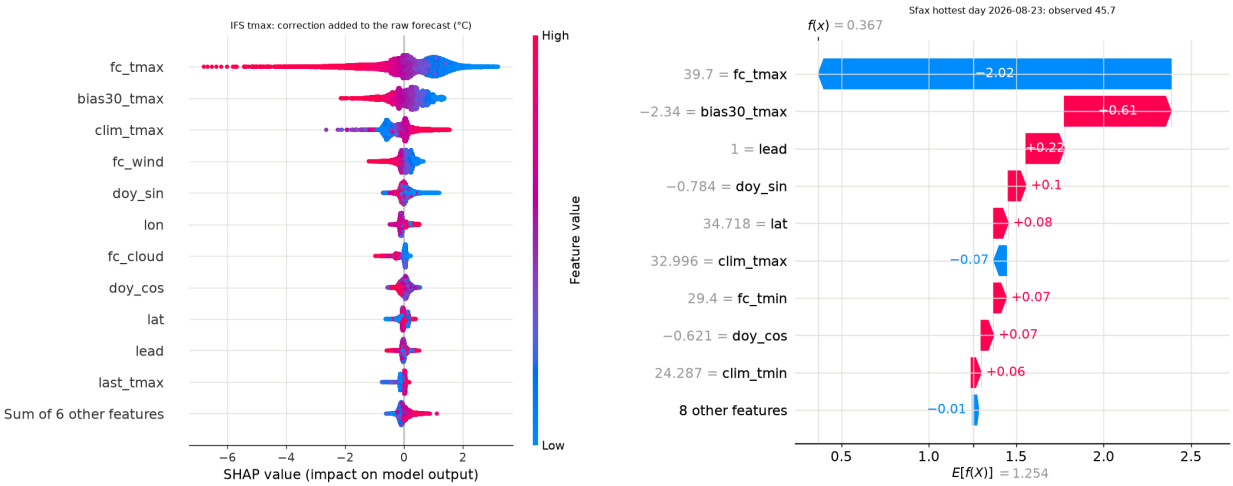


Figure 3: Left: SHAP values of the IFS $T_{\max}$ correction in the test period. Right: contributions for Sfax on 2026-08-23 at lead 1 (base value 1.25 °C). Predictor names are as in the code: `fc_` forecast, `bias30_` 30-day mean error, `clim_` climatology, `doy` day of year.

4.6 Drought outlook

As a side experiment we forecast SPI-3 (McKee et al., 1993) 1–3 months ahead from ERA5 monthly rain (Hersbach et al., 2020) from 1991 onwards. The predictors are the current SPI-1, SPI-3 and SPI-6, the month and the NAO index; the model is retrained every year and tested on 2011 onwards. The MAE is 0.61 at one month (persistence 0.66, climatology 0.81) and 0.69 at two months (0.93 and 0.81). At three months it is worse than climatology (0.88 against 0.81). The model also detects fewer drought months (SPI < -1) than persistence, 31% against 47% at one month.

5 Limitations

The test period covers only 14 months, and AIFS has the shortest archive, so its first models are trained on fewer rows. Open-Meteo keeps one deterministic run per day, so ensemble information is not used. The LightGBM settings were not tuned, so the gap to linear MOS may be slightly underestimated. Many comparisons are reported without correction for multiple testing, which matters for the small differences (LightGBM against linear MOS) but not for the large ones (corrected against raw). Finally, the comparison of the three models is only for corrected daily extremes at these 12 stations.

6 Conclusion

At these 12 stations a local correction removes 30–50% of the temperature error of IFS, AIFS and GFS at short leads, and turns rain forecasts that are worse than climatology into calibrated probabilities. To answer the question in the title: AIFS does need post-processing at these stations, since its raw errors are as large as those of IFS. But a simple correction is enough. Its errors are mostly a stable offset, so a running bias correction or linear MOS captures nearly all of the gain, and after that AIFS gives the lowest error of the three models. For IFS and GFS, LightGBM adds a small but significant improvement over linear MOS. The quantile intervals need conformal recalibration to reach their nominal coverage. Heavy rain and record heat are the main weaknesses. Natural next steps are ensemble input (ECMWF ENS, AIFS ENS) and a longer AIFS archive.

Code and data availability

Code, tests and result tables: `https://github.com/KiraiEEE/Weather`. The data are downloaded by the scripts from NOAA, Ogimet and Open-Meteo.

Acknowledgements

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